

Module-1

Time Value of Money

Key Concepts and Skills

- Be able to compute the future value and/or present value of a single cash flow or series of cash flows
- Be able to compute the return on an investment
- Understand perpetuities and annuities

The One-Period Case

- If you were to invest \$10,000 at 5-percent interest for one year, your investment would grow to \$10,500.

\$500 would be interest ($\$10,000 \times .05$)

\$10,000 is the principal repayment ($\$10,000 \times 1$)

\$10,500 is the total due. It can be calculated as:

$$\$10,500 = \$10,000 \times (1.05)$$

- The total amount due at the end of the investment is call the *Future Value (FV)*.

Future Value

- In the one-period case, the formula for FV can be written as:

$$FV = C_0 \times (1 + r)$$

Where C_0 is cash flow today (time zero), and r is the appropriate interest rate.

Present Value

- If you were to be promised \$10,000 due in one year when interest rates are 5-percent, your investment would be worth \$9,523.81 in today's dollars.

$$\$9,523.81 = \frac{\$10,000}{1.05}$$

The amount that a borrower would need to set aside today to be able to meet the promised payment of \$10,000 in one year is called the *Present Value (PV)*.

Note that $\$10,000 = \$9,523.81 \times (1.05)$.

Present Value

- In the one-period case, the formula for PV can be written as:

$$PV = \frac{C_1}{1 + r}$$

Where C_1 is cash flow at date 1, and
 r is the appropriate interest rate.

Net Present Value

- The Net Present Value (*NPV*) of an investment is the present value of the expected cash flows, less the cost of the investment.
- Suppose an investment that promises to pay \$10,000 in one year is offered for sale for \$9,500. Your interest rate is 5%. Should you buy?

Net Present Value

$$NPV = -\$9,500 + \frac{\$10,000}{1.05}$$

$$NPV = -\$9,500 + \$9,523.81$$

$$NPV = \$23.81$$

The present value of the cash inflow is greater than the cost. In other words, the Net Present Value is positive, so the investment should be purchased.

Net Present Value

In the one-period case, the formula for *NPV* can be written as:

$$NPV = -Cost + PV$$

If we had *not* undertaken the positive *NPV* project considered on the last slide, and instead invested our \$9,500 elsewhere at 5 percent, our *FV* would be less than the \$10,000 the investment promised, and we would be worse off in *FV* terms :

$$\$9,500 \times (1.05) = \$9,975 < \$10,000$$

The Multiperiod Case

1. ■ The general formula for the future value of an investment over many periods can be written as:

$$FV = C_0 \times (1 + r)^T$$

Where

2. C_0 is cash flow at date 0,

r is the appropriate interest rate, and

T is the number of periods over which the cash is invested.

Future Value

- Suppose a stock currently pays a dividend of \$1.10, which is expected to grow at 40% per year for the next five years.
- What will the dividend be in five years?

$$FV = C_0 \times (1 + r)^T$$

$$\$5.92 = \$1.10 \times (1.40)^5$$

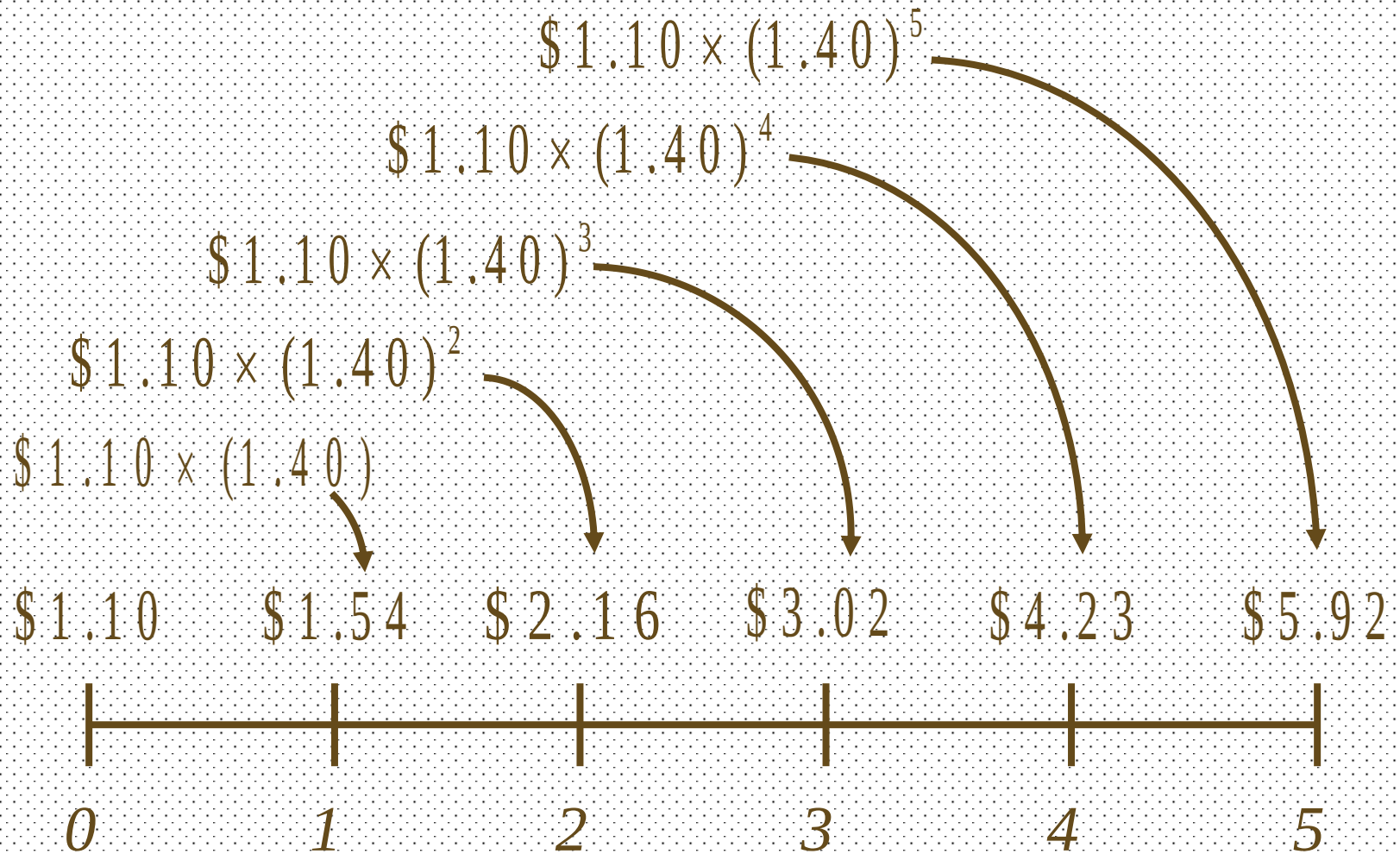
Future Value and Compounding

- Notice that the dividend in year five, \$5.92, is considerably higher than the sum of the original dividend plus five increases of 40-percent on the original \$1.10 dividend:

$$\$5.92 > \$1.10 + 5 \times [\$1.10 \times .40] = \$3.30$$

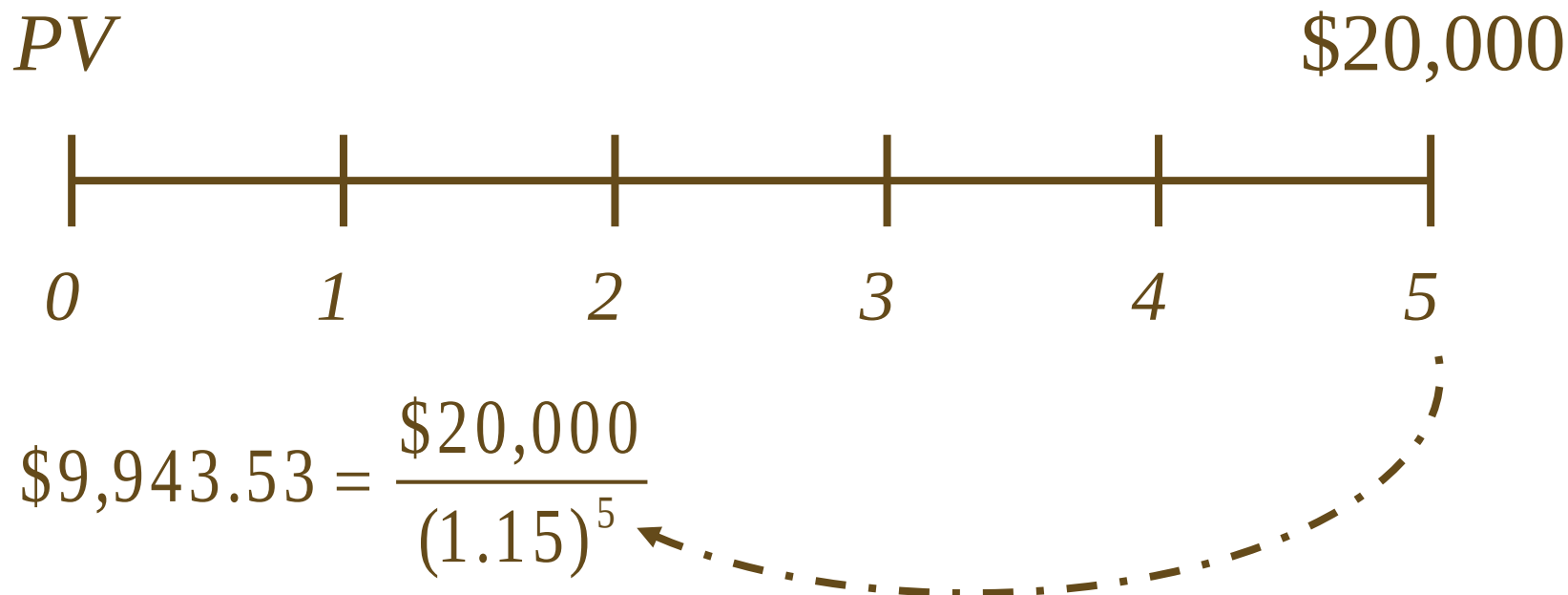
This is due to *compounding*.

Future Value and Compounding



Present Value and Discounting

- How much would an investor have to set aside today in order to have \$20,000 five years from now if the current rate is 15%?



Finding the Number of Periods

If we deposit \$5,000 today in an account paying 10%, how long does it take to grow to \$10,000?

$$FV = C_0 \times (1 + r)^T \qquad \$10,000 = \$5,000 \times (1.10)^T$$

$$(1.10)^T = \frac{\$10,000}{\$5,000} = 2$$

$$\ln(1.10)^T = \ln(2)$$

$$T = \frac{\ln(2)}{\ln(1.10)} = \frac{0.6931}{0.0953} = 7.27 \text{ years}$$

What Rate Is Enough?

Assume the total cost of a college education will be \$50,000 when your child enters college in 12 years. You have \$5,000 to invest today. What rate of interest must you earn on your investment to cover the cost of your child's education?

About 21.15%.

$$FV = C_0 \times (1 + r)^T \qquad \$50,000 = \$5,000 \times (1 + r)^{12}$$

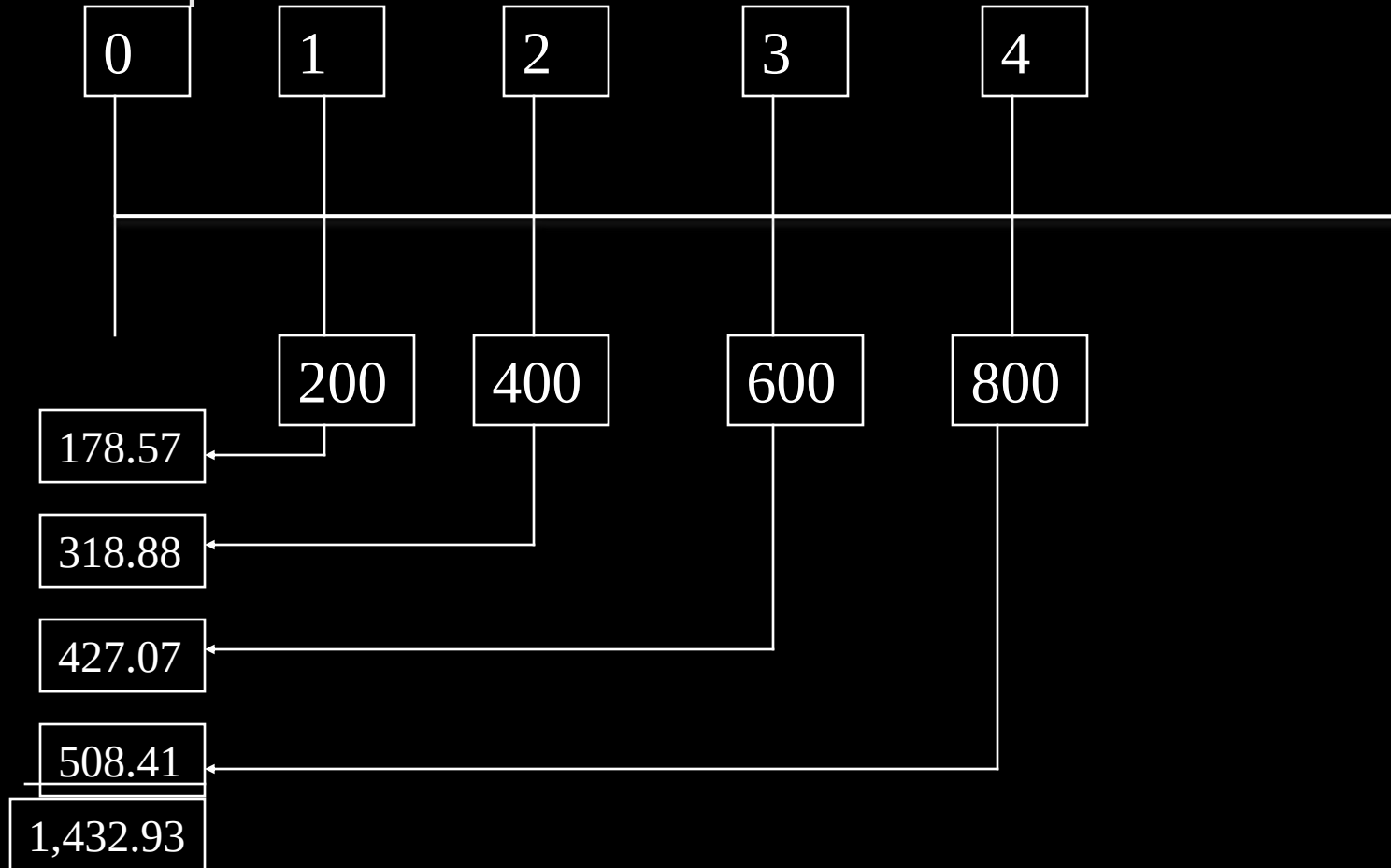
$$(1 + r)^{12} = \frac{\$50,000}{\$5,000} = 10 \qquad (1 + r) = 10^{1/12}$$

$$r = 10^{1/12} - 1 = 1.2115 - 1 = .2115$$

Multiple Cash Flows

- Consider an investment that pays \$200 one year from now, with cash flows increasing by \$200 per year through year 4. If the interest rate is 12%, what is the present value of this stream of cash flows?
- If the issuer offers this investment for \$1,500, should you purchase it?

Multiple Cash Flows



Present Value < Cost $\rightarrow$ Do Not Purchase

Compounding Periods

Compounding an investment m times a year for T years provides for future value of wealth:

$$FV = C_0 \times \left(1 + \frac{r}{m}\right)^{m \times T}$$

Compounding Periods

- For example, if you invest \$50 for 3 years at 12% compounded semi-annually, your investment will grow to:

$$FV = \$50 \times \left(1 + \frac{.12}{2}\right)^{2 \times 3} = \$50 \times (1.06)^6 = \$70.93$$

Continuous Compounding

- The general formula for the future value of an investment compounded continuously over many periods can be written as:

$$FV = C_0 \times e^{rT}$$

Where

C_0 is cash flow at date 0,

r is the stated annual interest rate,

T is the number of years, and

e is a transcendental number approximately equal to 2.718. e^x is a key on your calculator.

Simplifications

► Perpetuity

- A constant stream of cash flows that lasts forever

► Growing perpetuity

- A stream of cash flows that grows at a constant rate forever

► Annuity

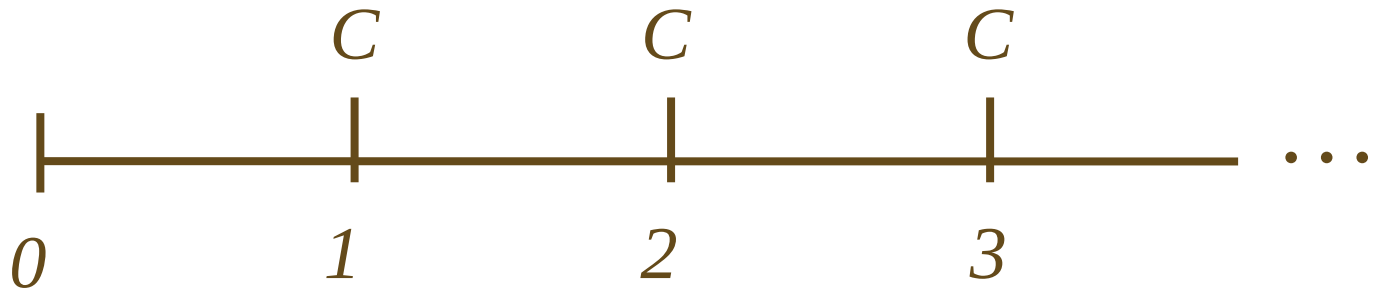
- A stream of constant cash flows that lasts for a fixed number of periods

► Growing annuity

- A stream of cash flows that grows at a constant rate for a fixed number of periods

Perpetuity

A constant stream of cash flows that lasts forever



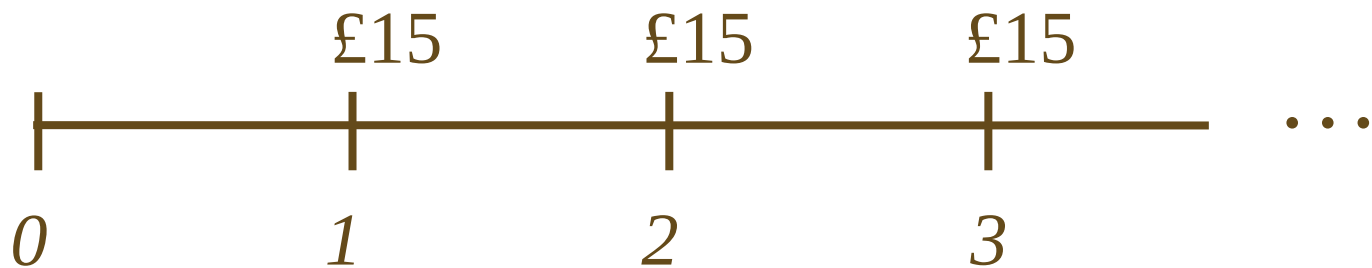
$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots$$

$$PV = \frac{C}{r}$$

Perpetuity: Example

What is the value of a British consol that promises to pay £15 every year for ever?

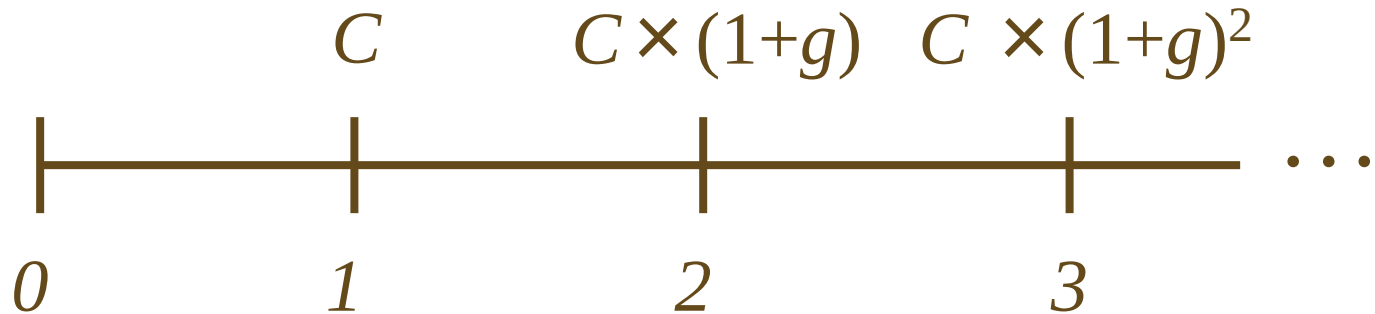
The interest rate is 10-percent.



$$PV = \frac{£15}{.10} = £150$$

Growing Perpetuity

A growing stream of cash flows that lasts forever



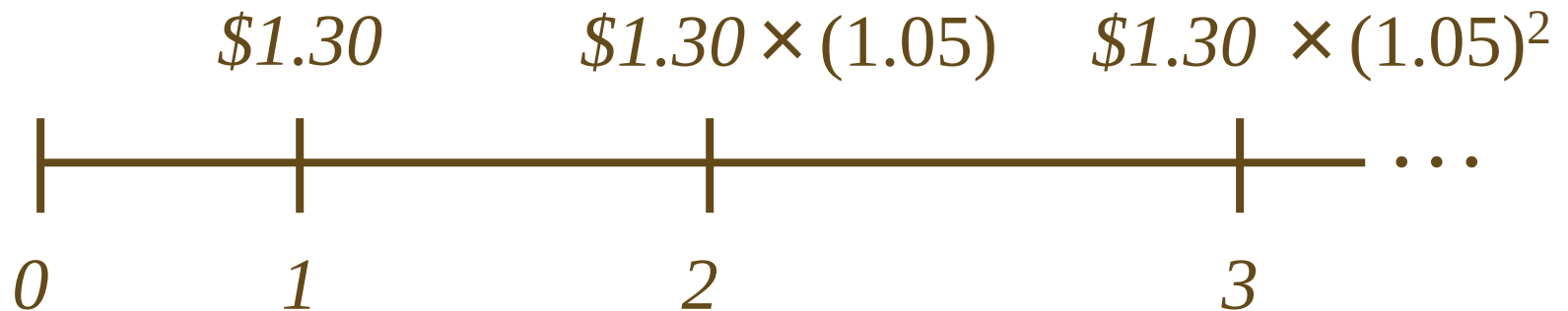
$$PV = \frac{C}{(1+r)} + \frac{C \times (1+g)}{(1+r)^2} + \frac{C \times (1+g)^2}{(1+r)^3} + \dots$$

$$PV = \frac{C}{r-g}$$

Growing Perpetuity: Example

The expected dividend next year is \$1.30, and dividends are expected to grow at 5% forever.

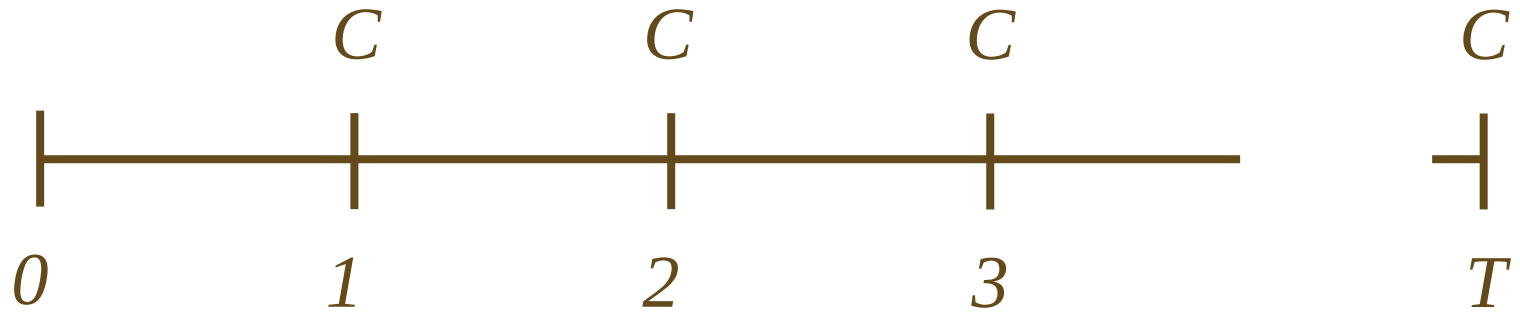
If the discount rate is 10%, what is the value of this promised dividend stream?



$$PV = \frac{\$1.30}{.10 - .05} = \$26.00$$

Annuity

A constant stream of cash flows with a fixed maturity

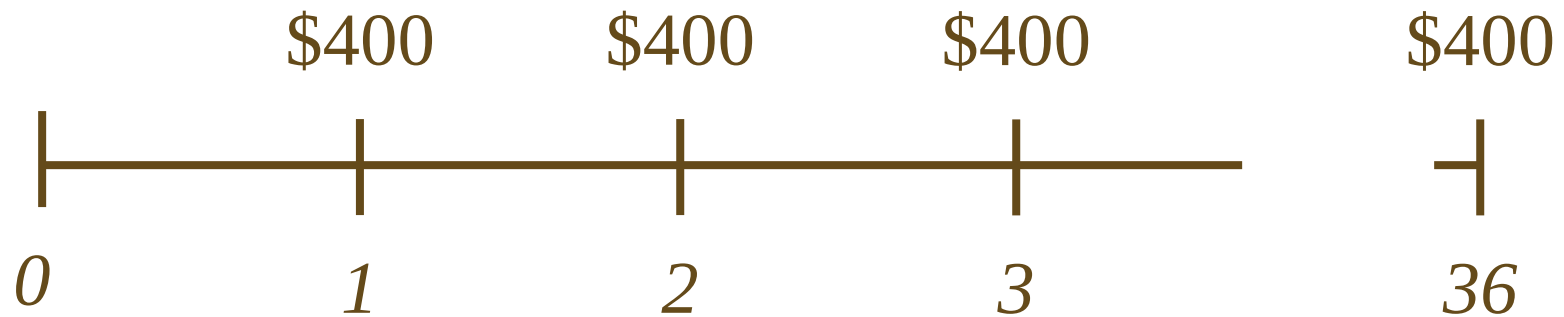


$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots + \frac{C}{(1+r)^T}$$

$$PV = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

Annuity: Example

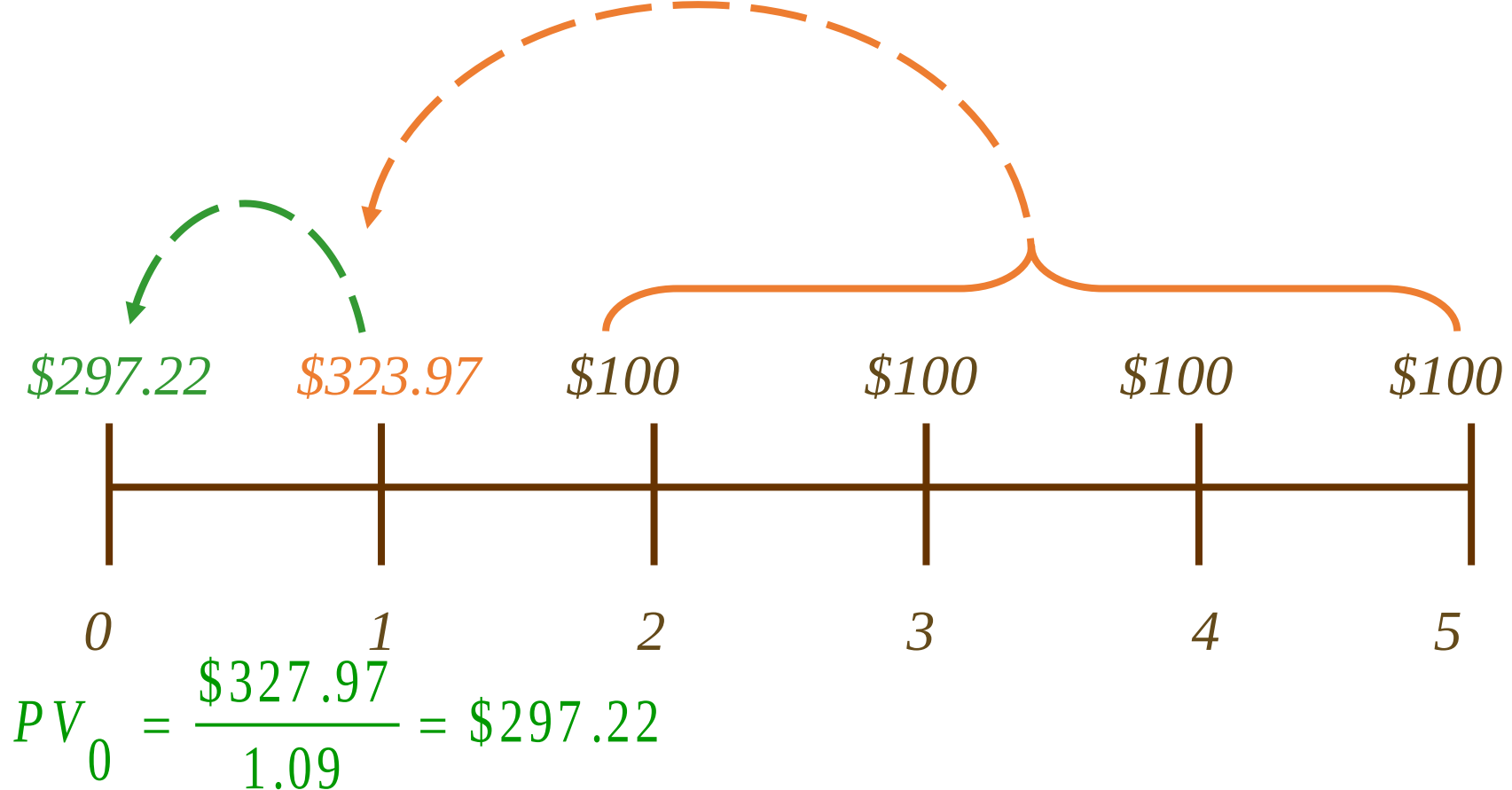
If you can afford a \$400 monthly car payment, how much car can you afford if interest rates are 7% on 36-month loans?



$$PV = \frac{\$400}{.07 / 12} \left[1 - \frac{1}{(1 + .07 / 12)^{36}} \right] = \$12,954.59$$

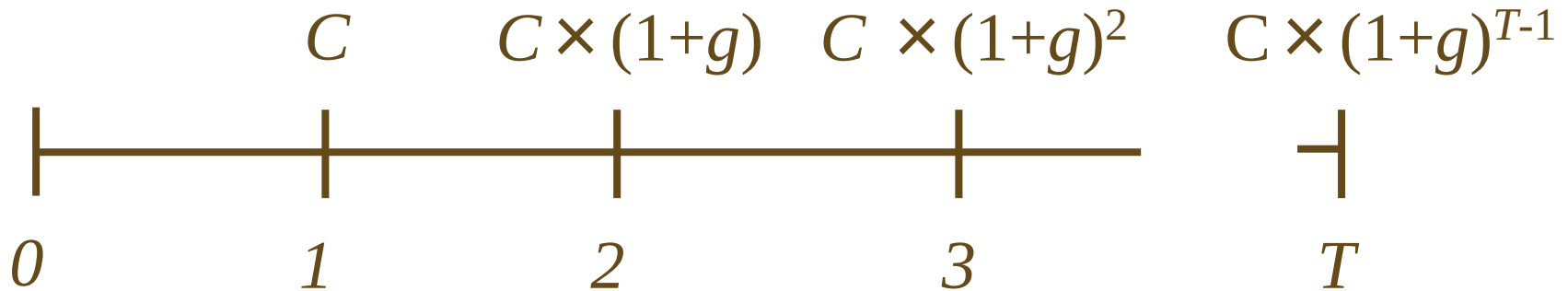
What is the present value of a four-year annuity of \$100 per year that makes its first payment two years from today if the discount rate is 9%?

$$PV_1 = \sum_{t=1}^4 \frac{\$100}{(1.09)^t} = \frac{\$100}{(1.09)^1} + \frac{\$100}{(1.09)^2} + \frac{\$100}{(1.09)^3} + \frac{\$100}{(1.09)^4} = \$323.97$$



Growing Annuity

A growing stream of cash flows with a fixed maturity

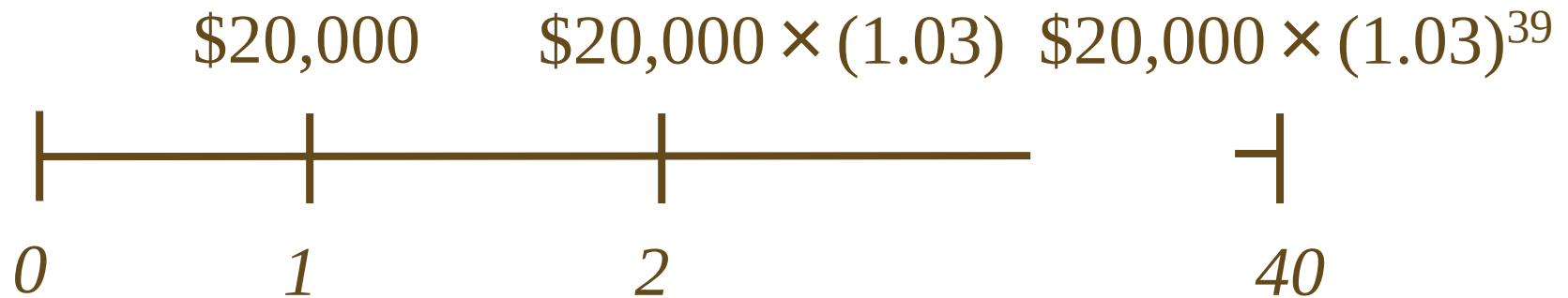


$$PV = \frac{C}{(1+r)} + \frac{C \times (1+g)}{(1+r)^2} + \dots + \frac{C \times (1+g)^{T-1}}{(1+r)^T}$$

$$PV = \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right]$$

Growing Annuity: Example

A defined-benefit retirement plan offers to pay \$20,000 per year for 40 years and increase the annual payment by 3% each year. What is the present value at retirement if the discount rate is 10%?



$$PV = \frac{\$20,000}{.10 - .03} \left[1 - \left(\frac{1.03}{1.10} \right)^{40} \right] = \$265,121.57$$

What Is a Firm Worth?

- Conceptually, a firm should be worth the present value of the firm's cash flows.
- The tricky part is determining the size, timing, and *risk* of those cash flows.

Disclaimer: This study material given in the slides have been taken from Ross, Westerfield, Jaffe, Jordan (TMH, 11e) and Corporate Finance: Theory and Practice (2nd ed.) authored by A. Damodaran for the classroom discussion at BITS Pilani only.